

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

01 / PROBLEM

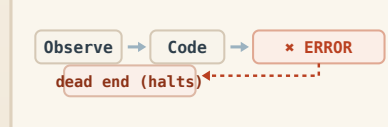
Why current agents fail.

Sustained autonomous research breaks in three different ways.

01

Closed-world fragility

Fixed workflows shatter on unexpected runtime errors.



Unhandled runtime faults halt unassisted runs

02

Trajectory momentum

A single context carries past momentum into local basins.



22 runs trapped tweaking parameters in same basin

03

Context inflation cost

Reconstructing context costs far more than the edit itself.



10 files read to change 1 line; burns token budget

DESIGN RESPONSE

Keep the world. Reset the trajectory.

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP

Persistent world · ephemeral trajectory · audited write-back

Fresh Scientist

Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

META Review

Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE

Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · EVIDENCE

Knowledge + Ledger

Metrics, failures, reports, and experiment receipts in **ledger.jsonl**.

B · PRIORS

Brief + Hypotheses

Current understanding and exploration leads in **brief.md**, open to revision.

C · STATE

Code + State

Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET

The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

WORKTREE ISOLATION

Independent git worktrees isolate branches and preserve reproducible artifacts.

SELF-HEALING

Traceback → patch → rerun keeps autonomous research moving after runtime faults.

PARALLEL SYNTHESIS

Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not prior reasoning momentum.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000

Official FM baseline

+0.0043363 Primary improvement

GAUC 0.6728421

(+0.0054)

nDCG@5 0.5390304

(+0.0033)

LogLoss 0.38102

(-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms

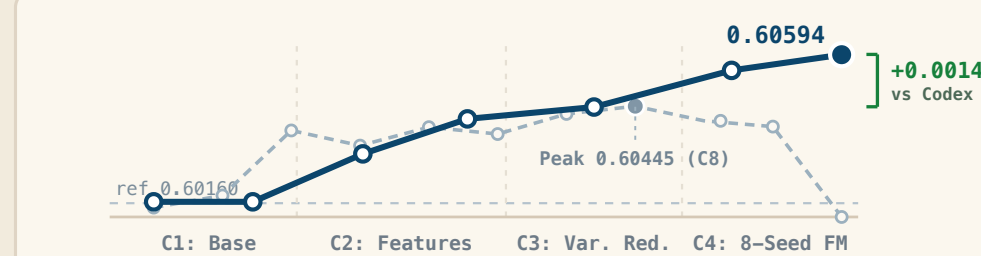
Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); 100% CPU inference without GPU clusters.

02 SUSTAINED SEARCH

The search stayed productive.

SciOdyssey · 4 cycles Codex Goal · 11 cycles



4 cycles 7 Full evaluations E003-E013 frontier +0.00148 over Codex

7 Full public-validation evaluations retained across 4 cycles. Trajectory climbed via **structural feature expansion** rather than single lucky edits.

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING CRASH AUTO-RECOVERY
FAILURE Cycle 1 numpy.float32 serialization crash
ACTION Scientist parsed traceback, auto-patched scalar helper
OUTCOME 0 human bugfixes · Unattended rerun succeeded

TIER 2 · META GATEKEEPING SPURIOUS STATE REJECTION
FALSE CLAIM Scientist claimed BPR gain on medium proxy
ACTION META audited system/data.py, caught UNK leak
OUTCOME Promotion blocked · Zero contaminated state

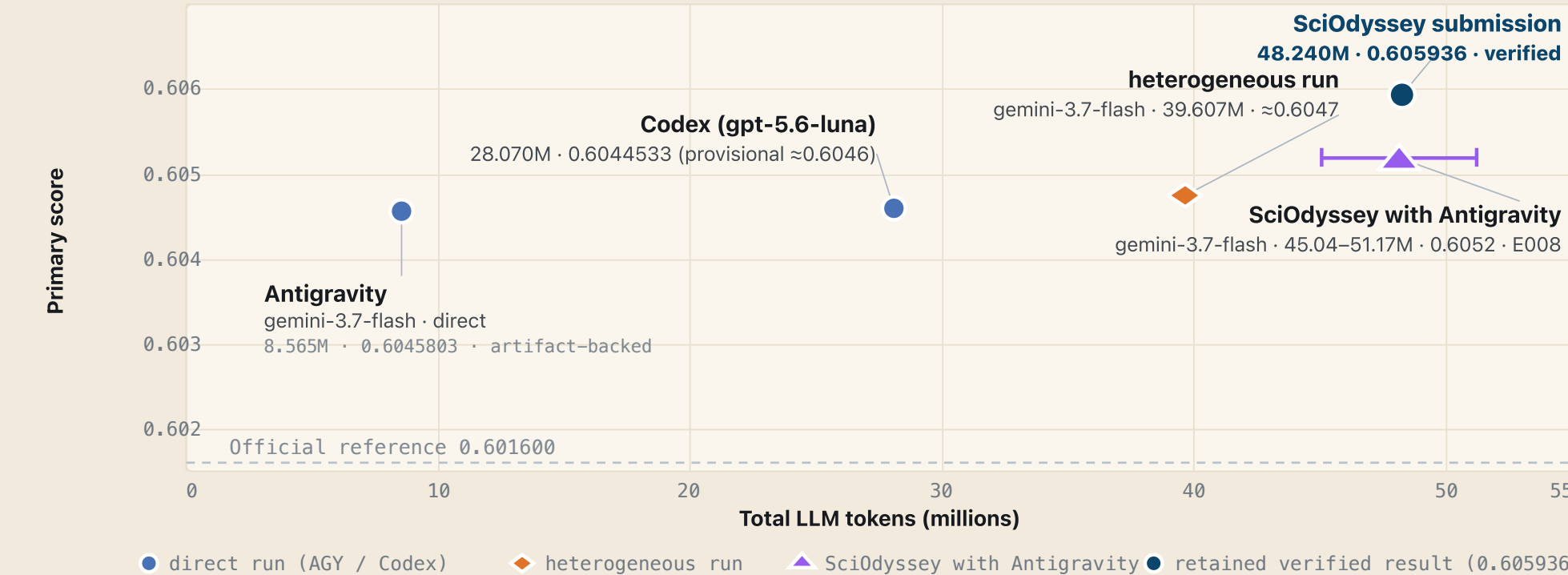
Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

05 RESOURCE ACCOUNTING

We are not tokenmaxxing. Score vs token investment.

~\$10 cost 1 × M2 0 GPU-hr 1h 55m

KuaiRand-Pure · Public validation · Inclusive input+output tokens



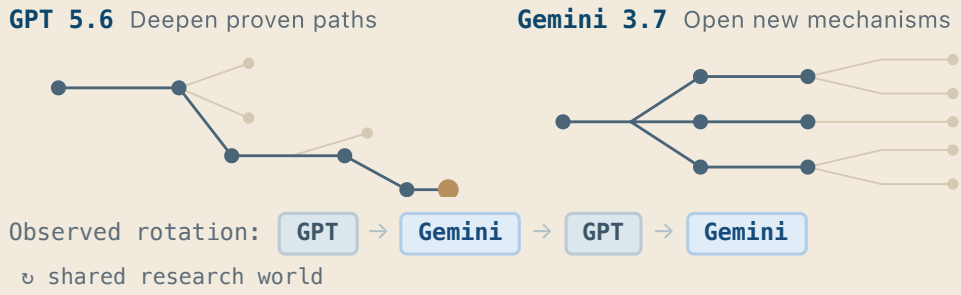
All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / FINDINGS

What the run revealed.

01 · MODEL DIVERSITY

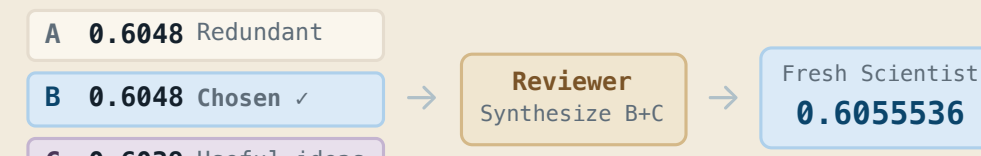
Model diversity can help break plateaus.



Gemini-only 0.6052 → GPT+Gemini 0.6059363

02 · PARALLEL SYNTHESIS

A losing branch can still improve the winner.



Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.

Branch B+C Ideas → Synthesis 0.605536

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS

Tier 1 Unattended Overnight Runs · Tier 2 Checkpoint Review (META) · Tier 3 Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

Autonomous by default.

Supervisable by design.

